

Kunal Kaushik

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EDUCATION

Bachelor of Science, Computer Science | Vanderbilt University | Nashville, TN

May 2023

- Cumulative GPA: 3.54 / 4.0
- Relevant Coursework: Algorithms, Artificial Intelligence, Data Structures, Linear Algebra, Operating Systems, Software Design & Patterns, Web-based Architecture, Undergraduate Research, Differential Equations
- Previous major: Civil Engineering (Freshman & Sophomore) — completed coursework; performed lab research in structural systems and soil mechanics before switching to CS.

RESEARCH EXPERIENCE

Independent Researcher | NashAlert — Civic Infrastructure AI Project

March 2026 – Present

- Independently designing and building NashAlert, a civic infrastructure reporting and prioritization system for Nashville, TN, investigating how open civic data can improve infrastructure maintenance prioritization and surface equity disparities in municipal service delivery.
- Ingested and cleaned Nashville's full 311 open dataset (334,710 infrastructure complaints, 2017–2026) into a PostGIS-enabled Supabase database with custom ETL pipeline handling coordinate normalization, category mapping, and EWKT spatial encoding.
- Designed and implemented a spatiotemporal recurrence scoring engine that computes composite infrastructure risk scores across 30,979 geospatial grid points using exponential decay weighting, severity-weighted complaint categorization, and resolution time analysis.
- Built a city planner dashboard (React + Mapbox GL JS) with a heatmap visualization of recurrence scores, a filtered priority queue, and a temporal complaint activity chart for each location.
- Currently extending with statistical anomaly detection (z-score-based complaint surge detection against historical seasonal baselines) and spatiotemporal KDE-based prediction of infrastructure hotspots 30–90 days ahead.
- Core research questions: (1) Can community-sourced reports combined with open civic data improve infrastructure maintenance prioritization? (2) Do lower-income Nashville neighborhoods bear a disproportionate infrastructure burden that a complaint-volume-driven system would fail to surface?

Undergraduate Research Assistant | LocalLens — NSF-Sponsored Platform | Vanderbilt University, Nashville, TN

Spring 2023 — CS 3860 under Dr. Daniel Balasubramanian, ISIS Lab

- Contributed to LocalLens, a civic platform helping Nashville communities report and locate lost items and animals using open municipal data.
- Developed frontend components and owned Android-side testing and debugging for the LocalLens Android app.
- Took part in team discussions shaping feature scope, architecture, and other product decisions throughout development.

Undergraduate Research Assistant | LASIR Lab — Structural Health Monitoring | Vanderbilt University, Nashville, TN

Fall 2017 — ME 3860 under Dr. Douglas Adams

- Assisted in experimental research on wind turbine structural health monitoring in the Laboratory for Systems Integrity and Reliability (LASIR).
- Supported data collection and analysis for vibration-based fault detection experiments on scaled wind turbine models.
- Exposure to physical sensing systems, signal analysis, and the challenges of infrastructure monitoring at scale.

Undergraduate Research Assistant | Soil Analysis Lab | Vanderbilt University, Nashville, TN

Spring 2017 — under Dr. David Kosson

- Conducted soil sample preparation, analysis, and data recording in an environmental civil engineering research lab for soil samples from Sterling Ranch, Colorado, focused on contaminant transport and material characterization.
- Developed foundational research skills: experimental protocol, data collection, and laboratory documentation.

PROFESSIONAL EXPERIENCE

Founder & Lead Engineer | Kosha — AI-Augmented Industrial Discovery Platform | Sussex, NJ

October 2025 – Present | FastAPI, PostgreSQL/PostGIS, Next.js, Gemini 3 Flash, Exa API

- Designed a full-stack B2B industrial discovery platform across 175+ Python modules with a pluggable search-provider abstraction (ABC pattern), layered async pipelines, and Alembic-versioned schema evolution.
- Built a two-stage AI lead pipeline — Exa search + content fetch followed by offline LLM normalization — deliberately decoupling browsing from reasoning to prevent hallucination propagation into the database.
- Designed a first-class provenance model attaching source, discovery provider, query, evidence URLs, and confidence score to every data assertion — with hard downgrade protection on merges for full auditability.
- Implemented a rule-based lead graduation system promoting AI-generated candidates to verified company records only when independently confirmed via HTTP 200, geocoded address, or valid E.164 phone.
- Built multi-signal geospatial search over PostGIS composing relevance, geodesic distance (ST_DWithin), and verified status; sponsored results capped algorithmically at ≤ 1 per 5 organic to preserve ranking integrity.

Software Developer | EquiLend Holdings LLC | Berkeley Heights, NJ

July 2023 – September 2025

- Designed and deployed an AI-powered automated code review pipeline (Python, AWS Bedrock/Claude, Gerrit REST API) with diff chunking for prompt optimization, incremental hash caching eliminating 80% of redundant reviews, and threaded comment handling — reducing CI review time by 70%.
- Led development of an Order Management Service microservice (Java, Spring Boot, PL/SQL) for securities lending with validation, tracking, and error recovery for high-volume institutional order management.
- Built a customizable Python CLI for real-time security availability queries and trade execution via SPIRE APIs, reducing client reliance on the web interface.
- Refactored PL/SQL queries for trade booking and reporting modules, achieving a 6.5-minute reduction in average query latency per client session.

Software Engineer Intern | Meta | New York, NY

May 2022 – August 2022

- Rebuilt four core Content Preferences screens in the Facebook Mobile App (iOS & Android) using React Native, PHP/Hack, GraphQL, and Meta's internal Bloks UI system — now live for millions of users globally.
- Integrated user interaction telemetry to track engagement with personalization flows, informing UX iteration across platforms.

Grove Collaborative | San Francisco, CA

DevOps Engineer Intern — January 2022 – May 2022

- Migrated CI/CD from Jenkins to GitHub Actions and upgraded Kubernetes deployments from Flux v1 to v2 with zero production downtime.
- Implemented memory/resource limit recommendations across Helm charts, improving pod stability by 22% in staging.

Software Engineer Intern — May 2021 – August 2021

- Improved checkout experience and subscription management APIs using Django REST, Python, Vue.js, and JavaScript.

SKILLS

Languages: Python, TypeScript, Java, JavaScript, SQL (PL/SQL)

Frontend & Mobile: React, Next.js, React Native/Expo

Backend & APIs: FastAPI, Spring Boot, Node/Express, REST, GraphQL

Data & Infrastructure: PostgreSQL/PostGIS, Supabase, OracleDB, Docker, GitHub Actions

AI & Machine Learning: LLM applications (AWS Bedrock/Claude, Gemini, OpenAI), prompt engineering; Python scientific stack (numpy, scipy, scikit-learn, pandas, geopandas)

Applied Statistics & Geospatial Analysis: spatiotemporal modeling, kernel density estimation, anomaly detection, geospatial risk scoring